# Assessment Menu: Teacher Guide

The Line Cook Bistro · design notes, grading and answer key

**Why students cannot just pick the easy ones**

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

| Skill group | What it covers | Dishes |
|---|---|---|
| **WRITE** | Writing equations of lines (slope-intercept, point-slope, standard), parallel and perpendicular | A1, A2, E1, E6, S1 |
| **MODEL** | Linear models, rate of change and lines of best fit | A3, E3, S2 |
| **SYSTEMS** | Solving systems of two and three equations | A4, E4, E5, S3 |
| **REGION** | Systems of inequalities and linear programming | A5, A6, E2, S4 |

| If a student only does… | Maximum |
|---|---|
| Only Appetizers (4 × 5) | 20 |
| Appetizers + Entrées (4 × 5 + 4 × 10) | 60 |
| Full meal (adds 2 Chef's Specials) | 100 |

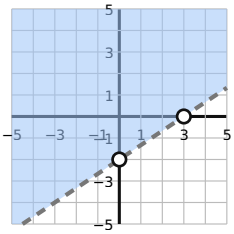
**Grading and options**

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

| Student | Appetizers (of 20) | Entrées (of 40) | Specials (of 40) | Skills OK? | Total |
|---|---|---|---|---|---|
| | | | | | |
| | | | | | |

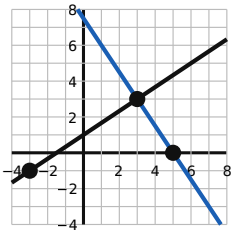
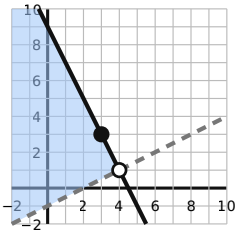
# Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Answer graphs are drawn from the functions themselves.

| Dish | Problem | Solution | Rubric |
|---|---|---|---|
| **A1** WRITE | A linear function $f(x)$ has $f(-2) = 5$ and $f(4) = -7$. **(a)** Find the slope. **(b)** Write the equation in point-slope form using $(-2, 5)$. **(c)** Write $f(x)$ in slope-intercept form. **(d)** Write it in standard form $Ax + By = C$. | **Answer:** (a) $-2$ (b) $y - 5 = -2(x + 2)$ (c) $f(x) = -2x + 1$ (d) $2x + y = 1$<br>(a) $m = \frac{-7-5}{4-(-2)} = \frac{-12}{6} = \mathbf{-2}$<br>(b) $\mathbf{y - 5 = -2(x + 2)}$ ($x - (-2)$ becomes $x + 2$).<br>(c) $y - 5 = -2x - 4$, so $\boldsymbol{f(x) = -2x + 1}$.<br>(d) Move the x term: $\mathbf{2x + y = 1}$. | 1 pts: (a) $m = -2$<br>1 pts: (b) Correct point-slope form with $x + 2$<br>1 pts: (c) $f(x) = -2x + 1$<br>1 pts: (d) Moves the x term to the left side<br>1 pts: (d) $2x + y = 1$ |
| **A2** WRITE | Line $\ell$ is $3x - 4y = 8$. **(a)** Find its slope. **(b)** Write the equation of the line through $(8, -3)$ that is **parallel** to $\ell$. **(c)** Write the equation of the line through $(6, 1)$ that is **perpendicular** to $\ell$. Use slope-intercept form. | **Answer:** (a) $\frac{3}{4}$ (b) $y = \frac{3}{4}x - 9$ (c) $y = -\frac{4}{3}x + 9$<br>(a) $y = \frac{3}{4}x - 2$, so the slope is $\mathbf{\frac{3}{4}}$.<br>(b) Same slope: $y + 3 = \frac{3}{4}(x - 8)$, so $\mathbf{y = \frac{3}{4}x - 9}$.<br>(c) Slope $-\frac{4}{3}$: $y - 1 = -\frac{4}{3}(x - 6)$, so $\mathbf{y = -\frac{4}{3}x + 9}$. | 1 pts: (a) Slope $\frac{3}{4}$ after solving for $y$<br>2 pts: (b) Slope $\frac{3}{4}$ and passes through $(8, -3)$<br>1 pts: (c) Perpendicular slope $-\frac{4}{3}$ (opposite reciprocal)<br>1 pts: (c) Uses the point $(6, 1)$ |
| **A3** MODEL | A stockpot is draining. It holds 88 quarts after 2 minutes and 70 quarts after 5 minutes. **(a)** Find the rate of change, with units. **(b)** Write a linear model $V(t)$. **(c)** What does the V-intercept mean? **(d)** When are 40 quarts left? | **Answer:** (a) $-6$ quarts per minute (b) $V(t) = -6t + 100$ (c) it started with 100 quarts (d) $t = 10$ minutes<br>(a) $\frac{70-88}{5-2} = \mathbf{-6}$ **quarts per minute**.<br>(b) $V(t) = -6t + b$. Use $(2, 88)$: $88 = -12 + b$, so $b = 100$ and $\mathbf{V(t) = -6t + 100}$.<br>(c) $V(0) = 100$: the pot held **100 quarts at the start**.<br>(d) $-6t + 100 = 40$, so $-6t = -60$ and $\mathbf{t = 10}$ **minutes**. | 1 pts: (a) $-6$ quarts per minute (sign and units)<br>2 pts: (b) $V(t) = -6t + 100$<br>1 pts: (c) Starting amount, 100 quarts<br>1 pts: (d) $t = 10$ |
| **A4** SYSTEMS | Solve each system. Say whether it has one solution, no solution or infinitely many. **(a)** $y = 2x - 3$ and $4x - 2y = 6$ **(b)** $y = 3x + 1$ and $6x - 2y = 5$ **(c)** $x + y = 7$ and $3x - y = 5$ | **Answer:** (a) infinitely many (same line) (b) no solution (parallel lines) (c) (3, 4)<br>(a) $4x - 2(2x - 3) = 6$ gives $6 = 6$: **infinitely many** (same line).<br>(b) $6x - 2(3x + 1) = 5$ gives $-2 = 5$: **no solution** (parallel lines).<br>(c) Add: $4x = 12$, so $x = 3$, $y = 4$: **(3, 4)**. | 2 pts: (a) Reaches $6 = 6$ and concludes infinitely many (not no solution)<br>1 pts: (b) Reaches a false statement and concludes no solution<br>2 pts: (c) (3, 4) |
| **A5** REGION | Graph $2x - 3y < 6$ on the grid. **(a)** Should the boundary be solid or dashed? Why? **(b)** Shade the correct side and show the test point $(0, 0)$. | **Answer:** Dashed boundary through $(0, -2)$ and $(3, 0)$; shade the side containing $(0, 0)$, the side above the line. Intercepts of $2x - 3y = 6$: (3, 0) and (0, −2).<br>(a) Strict ($<$), so **dashed**.<br>(b) $0 < 6$ is true at $(0, 0)$, so shade **the side containing (0, 0)** (above the line).<br>  | 1 pts: Boundary line drawn through (3, 0) and (0, −2)<br>1 pts: Dashed because the inequality is strict<br>1 pts: Tests a point such as (0, 0)<br>2 pts: Shades the side containing (0, 0) |
| **A6** REGION | A system is $y > x - 2$ and $x + y \le 6$. Is each point a solution? **(a)** (1, 3) **(b)** (2, 5) **(c)** (4, 2) **(d)** (0, 0). In (c), the point satisfies one inequality; explain why it is still not a solution. | **Answer:** (a) yes (b) no (c) no (d) yes<br>(a) $3 > -1$ and $4 \le 6$: **yes**.<br>(b) $5 > 0$ but $7 \le 6$ is false: **no**.<br>(c) $6 \le 6$ is true, but $2 > 2$ is false (strict): **no**.<br>(d) $0 > -2$ and $0 \le 6$: **yes**. A solution must satisfy **every** inequality. | 1 pts: (a) yes<br>1 pts: (b) no (fails $x + y \le 6$)<br>1 pts: (c) no<br>1 pts: (d) yes<br>1 pts: (c) Explains that $2 > 2$ is false (strict) and all inequalities must hold |

# Answer Key: Entrées (10 pts each)

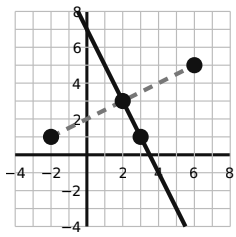
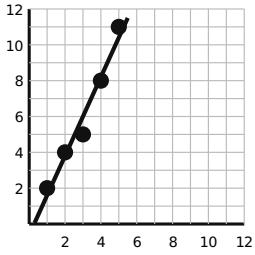
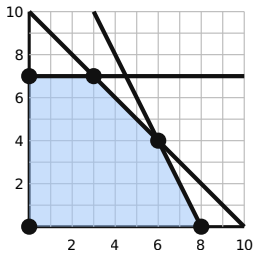
Worked solutions and rubrics. Answer graphs are drawn from the functions themselves.

| Dish | Problem | Solution | Rubric |
|---|---|---|---|
| **E1**<br>WRITE | Line $\ell$ passes through $(-3, -1)$ and $(3, 3)$. **(a)** Write $\ell$ in slope-intercept form. **(b)** Line m is perpendicular to $\ell$ and passes through $(3, 3)$. Write m in point-slope form and find its x-intercept. **(c)** Graph both lines on the grid and name their intersection. | **Answer:** (a) $y = \frac{2}{3}x + 1$ (b) $y - 3 = -\frac{3}{2}(x - 3)$; x-intercept $(5, 0)$ (c) both lines graphed; they meet at $(3, 3)$<br>  <br>(a) $m = \frac{3-(-1)}{3-(-3)} = \frac{4}{6} = \frac{2}{3}$. Then $y - 3 = \frac{2}{3}(x - 3)$ gives $\mathbf{y = \frac{2}{3}x + 1}$.<br>(b) Perpendicular slope: $-\frac{3}{2}$. Point-slope: $\mathbf{y - 3 = -\frac{3}{2}(x - 3)}$. Set $y = 0$: $x - 3 = 2$, so the x-intercept is **(5, 0)**.<br>(c) $\ell$: $(-3, -1)$, $(0, 1)$, $(3, 3)$. m: $(1, 6)$, $(3, 3)$, $(5, 0)$. They cross at **(3, 3)**. | 1 pts: (a) Slope $\frac{2}{3}$<br>2 pts: (a) $y = \frac{2}{3}x + 1$<br>2 pts: (b) Slope $-\frac{3}{2}$ (opposite reciprocal)<br>1 pts: (b) Point-slope form through (3, 3)<br>1 pts: (b) x-intercept (5, 0)<br>2 pts: (c) Both lines graphed accurately<br>1 pts: (c) Intersection (3, 3) |
| **E2**<br>REGION | Graph the system on the grid: $y > \frac{1}{2}x - 1$ and $2x + y \le 9$. **(a)** Which boundary is dashed? **(b)** Shade the solution region. **(c)** Find the point where the boundaries meet. Is it a solution? **(d)** Is (3, 3) a solution? | **Answer:** (a) $y = \frac{1}{2}x - 1$ is dashed (b) region below $2x + y = 9$ and above the dashed line (c) $(4, 1)$ is not a solution (d) yes<br>  <br>(a) $y > \ldots$ is strict: $\mathbf{y = \frac{1}{2}x - 1}$ **is dashed**; the other is solid.<br>(b) Test $(0, 0)$: above the dashed line and below $2x + y = 9$. Keep the overlap.<br>(c) $\frac{1}{2}x - 1 = 9 - 2x$ gives $\frac{5}{2}x = 10$, so $x = 4$ and $y = 1$. **(4, 1)** is on the dashed line ($1 > 1$ is false): **not** a solution.<br>(d) $3 > 0.5$ and $9 \le 9$: **yes** (solid line counts). | 1 pts: (a) Dashed line identified<br>2 pts: (b) Both boundaries drawn correctly (dashed and solid)<br>2 pts: (b) Shades the overlap only<br>2 pts: (c) Intersection (4, 1)<br>1 pts: (c) Not a solution because the dashed boundary is excluded<br>1 pts: (d) yes<br>1 pts: (d) Substitutes into both inequalities |
| **E3**<br>MODEL | A bike courier charges $C$(m) dollars for $m$ miles:<br>miles $m$: 2, 5, 9<br>cost $C$ ($): 11, 17, 25<br>**(a)** Show the data are linear. **(b)** Write $C$(m). **(c)** Interpret the slope and the C-intercept. **(d)** A rival charges $R$(m) = 2.5m + 4. For what m do they cost the same? Which is cheaper for longer trips? | **Answer:** (b) $C$(m) = 2m + 7 (c) \$2 per mile; \$7 base fee (d) m = 6 miles (\$19); the courier $C$(x) is cheaper for $m > 6$<br>(a) $\frac{17-11}{5-2} = 2$ and $\frac{25-17}{9-5} = 2$: the rate of change is constant, so the data are linear.<br>(b) $11 = 2(2) + b$, so $b = 7$: $\mathbf{C(m) = 2m + 7}$.<br>(c) **\$2 for each additional mile**; **\$7 base fee** (a 0-mile trip).<br>(d) $2m + 7 = 2.5m + 4$ gives $3 = 0.5m$, so $\mathbf{m = 6}$, where both cost \$19. For $m > 6$ the rival's steeper rate ($2.5 > 2$) costs more, so $C$(x) is cheaper. | 2 pts: (a) Computes both rates of change and finds them equal<br>2 pts: (b) C(m) = 2m + 7<br>1 pts: (c) Slope: \$2 per mile<br>1 pts: (c) Intercept: \$7 base fee<br>1 pts: (d) Sets 2m + 7 = 2.5m + 4<br>1 pts: (d) m = 6 (cost \$19)<br>2 pts: (d) Courier is cheaper for longer trips, with a reason (smaller slope) |

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|---|---|---|---|
| **E4**<br>SYSTEMS | Catering prices: 3 pasta trays and 2 salad trays cost \$96; 2 pasta trays and 5 salad trays cost \$141. **(a)** Define variables, write a system and solve it by elimination. **(b)** Making n trays costs C = 13n + 300 dollars and earns R = 25n dollars. How many trays to break even? What is the revenue then? | **Answer:** (a) pasta \$18 per tray, salad \$21 per tray (b) n = 25 trays, revenue \$625<br>(a) Let p = price of a pasta tray and s = price of a salad tray: 3p + 2s = 96 and 2p + 5s = 141. Multiply by 2 and 3: 6p + 4s = 192, 6p + 15s = 423. Subtract: 11s = 231, **s = 21**; then **p = 18**.<br>(b) Break even when 13n + 300 = 25n: 300 = 12n, so **n = 25** trays. Revenue: 25(25) = **\$625**. | 2 pts: (a) Variables defined and both equations written correctly<br>2 pts: (a) Elimination steps (matching coefficients, subtracting)<br>2 pts: (a) p = 18 and s = 21<br>1 pts: (b) Sets C = R<br>2 pts: (b) n = 25<br>1 pts: (b) \$625 |
| **E5**<br>SYSTEMS | **(a)** Solve by elimination:<br>$\begin{cases} x + y + z = 6 \\ 2x - y + z = 3 \\ x + 2y - z = 2 \end{cases}$ **(b)** Show that x + y + z = 4, 2x + 2y + 2z = 9, x − y = 1 has no solution. **(c)** A student solving another system reaches 0 = 0. What does that mean? | **Answer:** (a) (1, 2, 3) (b) 2(first) − (second) gives 0 = −1 (c) infinitely many solutions (dependent equations)<br>(a) Eq. 1 + Eq. 3: 2x + 3y = 8. Eq. 2 + Eq. 3: 3x + y = 5. Then **x = 1**, **y = 2**, and **z = 3**.<br>(b) 2(Eq. 1) − Eq. 2 gives 0 = −1: **no solution**.<br>(c) The equations are dependent: **infinitely many solutions**. | 2 pts: (a) Eliminates the same variable from two different pairs<br>2 pts: (a) Solves the resulting 2 by 2 system<br>2 pts: (a) (1, 2, 3) with z found by back-substitution<br>2 pts: (b) Reaches 0 = −1 and concludes no solution<br>2 pts: (c) Infinitely many solutions (dependent) |
| **E6**<br>WRITE | **(a)** The table shows a linear function.<br>$x$ \| 1 \| 3 \| 6<br>$f(x)$ \| 7 \| 13 \| 22<br>Show that it is linear and write $f(x)$. **(b)** Write $y - 1 = -\frac{2}{3}(x + 3)$ in standard form Ax + By = C with integer coefficients and A > 0. **(c)** Write the horizontal and the vertical line through (5, −2). | **Answer:** (a) $f(x) = 3x + 4$ (b) 2x + 3y = −3 (c) y = −2 and x = 5<br>(a) $\frac{13 - 7}{3 - 1} = 3$ and $\frac{22 - 13}{6 - 3} = 3$, so m = 3; 7 = 3(1) + b gives b = 4: ***f*(x) = 3x + 4**.<br>(b) Multiply by 3: 3y − 3 = −2(x + 3) = −2x − 6. Add 2x and 3: **2x + 3y = −3**.<br>(c) Horizontal: **y = −2**. Vertical: **x = 5**. | 2 pts: (a) Equal slopes computed from both intervals<br>2 pts: (a) f(x) = 3x + 4<br>2 pts: (b) Clears the fraction correctly (multiplies by 3)<br>2 pts: (b) 2x + 3y = −3<br>1 pts: (c) y = −2<br>1 pts: (c) x = 5 |

# Answer Key: Chef’s Specials (20 pts each)

Worked solutions and rubrics. Answer graphs are drawn from the functions themselves.

| Dish | Problem | Solution | Rubric |
|---|---|---|---|
| **S1** WRITE | A(−2, 1) and B(6, 5) are the endpoints of a segment. **(a)** Find its midpoint M. **(b)** Write the perpendicular bisector of AB in slope-intercept form. **(c)** Show that P(3, 1) is on the bisector and is equally far from A and B. **(d)** Graph A, B, M and the bisector on the grid, and find where the bisector crosses the x-axis. | **Answer:** (a) M(2, 3) (b) y = −2x + 7 (c) PA = PB = 5 (d) x-intercept ( $\frac{7}{2}$ , 0)<br>(a) M = **(2, 3)**<br>(b) Slope of AB = $\frac{4}{8} = \frac{1}{2}$, so the perpendicular slope is **−2**: y − 3 = −2(x − 2), **y = −2x + 7**.<br>(c) −2(3) + 7 = 1 ✓. PA = $\sqrt{(5^2 + 0^2)} = 5$, PB = $\sqrt{(3^2 + 4^2)} = 5$.<br>(d) y = 0 gives 2x = 7: x-intercept **( $\frac{7}{2}$ , 0)**. The line passes through (0, 7) and (3, 1).<br>  | 3 pts: (a) Midpoint (2, 3)<br>2 pts: (b) Slope of AB = $\frac{1}{2}$<br>2 pts: (b) Perpendicular slope −2<br>3 pts: (b) y = −2x + 7 (uses the midpoint)<br>2 pts: (c) Substitutes P into the line<br>3 pts: (c) PA = 5 and PB = 5 by the distance formula<br>2 pts: (d) A, B and M plotted<br>2 pts: (d) Bisector graphed<br>1 pts: (d) x-intercept ( $\frac{7}{2}$ , 0) |
| **S2** MODEL | A bakery’s online orders *y* in week *x* after launch: (1, 2), (2, 4), (3, 5), (4, 8), (5, 11). A calculator gives the line of best fit y = 2.2x − 0.6 with r = 0.98 (to the nearest hundredth). **(a)** Plot the points and graph the line. **(b)** Interpret the slope. **(c)** Predict weeks 7 and 20; which is less reliable? **(d)** Find the residual at week 4. | **Answer:** (b) about 2.2 more orders each week (c) week 7: 14.8 orders; week 20: 43.4 orders (week 20 is extrapolation) (d) 8 − 8.2 = −0.2<br>(a) Plot the points; the line passes through (1, 1.6) and (5, 10.4).<br>(b) Orders rise by about **2.2 per week** (r = 0.98: strong positive relationship).<br>(c) Week 7: 2.2(7) − 0.6 = **14.8** orders (inside the data). Week 20: 2.2(20) − 0.6 = **43.4** orders, far outside the data (extrapolation): **less reliable**.<br>  <br>(d) Predicted at week 4: 2.2(4) − 0.6 = 8.2. Residual = 8 − 8.2 = **−0.2** (actual − predicted). | 3 pts: (a) All five points plotted<br>3 pts: (a) Line y = 2.2x − 0.6 graphed correctly<br>4 pts: (b) 2.2 orders per week (rate, with units)<br>2 pts: (c) Week 7: 14.8 orders<br>2 pts: (c) Week 20: 43.4 orders<br>2 pts: (c) Week 20 is less reliable (extrapolation)<br>2 pts: (d) Predicted value 8.2<br>2 pts: (d) Residual −0.2 (actual minus predicted) |
| **S3** SYSTEMS | Receipts at the bistro (x = soup, y = salad, z = sandwich price): one of each costs $21; 2 soups, 1 salad and 3 sandwiches cost $44; 3 soups, 2 salads and 1 sandwich cost $38. **(a)** Write a system. **(b)** Solve it by elimination. **(c)** A receipt says 3 of each cost $60. Is that possible? **(d)** Find the cost of 4 soups, 3 salads and 2 sandwiches. | **Answer:** (a) x + y + z = 21, 2x + y + 3z = 44, 3x + 2y + z = 38 (b) soup $5, salad $7, sandwich $9 (c) no (d) $59<br>(a) x + y + z = 21, 2x + y + 3z = 44 and 3x + 2y + z = 38.<br>(b) Eq. 2 − Eq. 1: x + 2z = 23. Eq. 3 − Eq. 1: 2x + y = 17. Eliminate y again using Eq. 1 (y = 21 − x − z): x − z = −4. Subtract: 3z = 27, **z = 9**, x = 5, y = 7. Soup $5, salad $7, sandwich $9.<br>(c) 3(x + y + z) = 3(21) = 63, not 60: **impossible**.<br>(d) 4(5) + 3(7) + 2(9) = 20 + 21 + 18 = **$59**. | 1 pts: (a) Defines the three unknowns<br>3 pts: (a) Writes all three equations correctly<br>3 pts: (b) Eliminates one variable using two different pairs<br>3 pts: (b) Solves the resulting 2 by 2 system<br>3 pts: (b) Soup $5, salad $7, sandwich $9<br>4 pts: (c) No, with the reason 3 times 21 = 63<br>3 pts: (d) $59 |
| **S4** REGION | The bistro bakes x loaves and y pies a day. Oven hours: 2x + y ≤ 16. Prep hours: x + y ≤ 10. At most 7 pies. Profit: P = 4x + 5y dollars. **(a)** Graph the feasible region (x ≥ 0, y ≥ 0). **(b)** Find every vertex. **(c)** Find the maximum profit and where it occurs. **(d)** If P became 5x + 4y, which vertex would win? | **Answer:** (b) (0, 0), (8, 0), (6, 4), (3, 7), (0, 7) (c) $47 at 3 loaves and 7 pies (d) (6, 4), with $46<br>(a) Lines through (8, 0) and (6, 4); (3, 7) and (6, 4); y = 7. Shade inside all five constraints.<br>(b) (0, 0); (8, 0); (6, 4) and (3, 7) from pairs of boundaries; (0, 7).<br>(c) P: 0, 32, 44, **47**, 35. Maximum **$47 at (3, 7)**: 3 loaves, 7 pies.<br>(d) 5x + 4y: 0, 40, **46**, 43, 28, so **(6, 4)**.<br>  | 2 pts: (a) Line 2x + y = 16 correct<br>2 pts: (a) Line x + y = 10 correct<br>1 pts: (a) Line y = 7 correct<br>1 pts: (a) Correct region shaded<br>5 pts: (b) Five vertices, 1 point each<br>2 pts: (c) Evaluates P at every vertex<br>2 pts: (c) Maximum $47 at (3, 7)<br>1 pts: (c) Interprets in words (loaves and pies)<br>2 pts: (d) Re-evaluates with 5x + 4y<br>2 pts: (d) Vertex (6, 4) |